

Problem Statement

A values-led problem statement for detecting harmful intent before an LLM responds

Team	Komadugu Team
Challenge	Challenge 1 — Values-Led Project Statement
Related deliverables	data_card.pdf, impact_statement_card.pdf, stakeholder_engagement.pdf

1. What problem is this project trying to solve?

Large language models (LLMs) can sometimes respond to harmful or malicious user requests — prompts seeking instructions for illegal activity, violence, hate speech, or other unsafe content. Current safety mechanisms often rely on computationally expensive guardrails that operate during or after response generation, which makes them difficult to scale and less transparent. As AI systems are deployed more widely, there is a growing need for a faster, more efficient, and interpretable way to identify harmful user intent *before* a response is generated. Solving this improves the safety, reliability, and trustworthiness of AI systems while reducing the risk of harmful content reaching users.

2. Who is affected by this problem?

- Everyday users of AI systems who expect safe and trustworthy interactions.
- Organizations deploying LLM-based applications (e.g., customer support, education, healthcare, finance).
- AI developers and safety researchers responsible for preventing harmful model behavior.
- Online communities that may be exposed to toxic, abusive, or dangerous AI-generated content.
- Vulnerable groups disproportionately affected by hate speech, harassment, misinformation, or harmful instructions.

3. Why is solving this problem important?

If harmful or malicious prompts are not detected early, LLMs may generate toxic, dangerous, or misleading responses that can cause real-world harm — spreading misinformation, encouraging illegal or unsafe activity, reinforcing discrimination, or damaging users' trust in AI. Organizations may also face legal, ethical, and reputational risk if their products fail to prevent harmful interactions. Earlier detection of malicious intent makes AI systems safer, more reliable, and more trustworthy, while reducing the computational cost of relying solely on external, post-hoc safety mechanisms.

4. Which values guide the solution, and why?

Value	Why it was chosen
Safety	The project's core aim is reducing harmful AI outputs by catching intent before generation.
Fairness	Minimizes bias and prevents unjust treatment of particular users, languages, or communities.

Value	Why it was chosen
Transparency	Interpretable detection helps developers understand why harmful behavior is flagged, improving trust and debuggability.
Privacy	Protects sensitive user information during data processing.
Accountability	Encourages continuous monitoring and improvement of the system's decisions.
Reliability	Ensures consistent performance across different types of user input in real-world use.

5. What ethical risks should the team consider?

- False positives that incorrectly classify harmless prompts as toxic.
- False negatives that allow harmful prompts to pass undetected.
- Bias against certain languages, dialects, or cultural expressions.
- Privacy concerns when processing user inputs.
- Over-reliance on automated decisions without human oversight.
- Limited explainability if users cannot understand why their prompt was flagged.
- Potential misuse of probing techniques to analyze or exploit model internals.

6. How will the team reduce these risks?

- Evaluating the model on diverse datasets representing different cultures, languages, and communication styles.
- Measuring both false-positive and false-negative rates, not accuracy alone.
- Regularly auditing the system for bias and fairness.
- Using anonymized or privacy-preserving datasets whenever possible.
- Maintaining human oversight for high-risk or uncertain cases.
- Documenting model limitations and evaluation results transparently.
- Continuously testing and updating the probes as models and threats evolve.

7. What outcomes would define success?

- Accurate detection of harmful prompts with low false-positive and false-negative rates.
- Fair performance across diverse user groups and languages.
- Reduced computational cost compared to traditional safety-monitoring methods.
- Increased transparency into where safety-related information is encoded within the LLM.
- Improved trust in AI systems from developers and end users.
- A solution that could be integrated into real-world AI applications to enhance safety while preserving legitimate user interactions.

Final Values-Led Problem Statement

Large language models are increasingly used in education, healthcare, customer service, and other domains, yet they remain vulnerable to harmful or malicious user prompts that can lead to unsafe or toxic responses. This problem affects AI users, developers, organizations, and online communities — particularly vulnerable individuals who may be exposed to harmful content. Our project aims to improve the early detection of malicious intent by identifying safety-related patterns within a model's internal representations before harmful responses are generated. Our solution is guided by the values of safety, fairness, transparency, privacy, accountability, and reliability, to ensure that harmful behavior is detected without unfairly targeting legitimate users or compromising personal data. We hope to strengthen trust in AI systems by enabling more efficient, interpretable, and responsible safety monitoring that benefits both developers and the broader community.

8. From Statement to Implementation

This statement is operationalized as a linear probe trained on frozen Gemma-2-2B layer-14 hidden states, using the Civil Comments dataset as a proxy for harmful/toxic language patterns (see [data_card.pdf](#)). Its risks, mitigations, and success criteria are expanded in [impact_statement_card.pdf](#), and the plan for involving affected stakeholders as the project matures is detailed in [stakeholder_engagement.pdf](#).

Source: Komadugu Team, Challenge 1 — Values-Led Project Statement worksheet. Companion documents: [data_card.pdf](#), [impact_statement_card.pdf](#), [stakeholder_engagement.pdf](#).